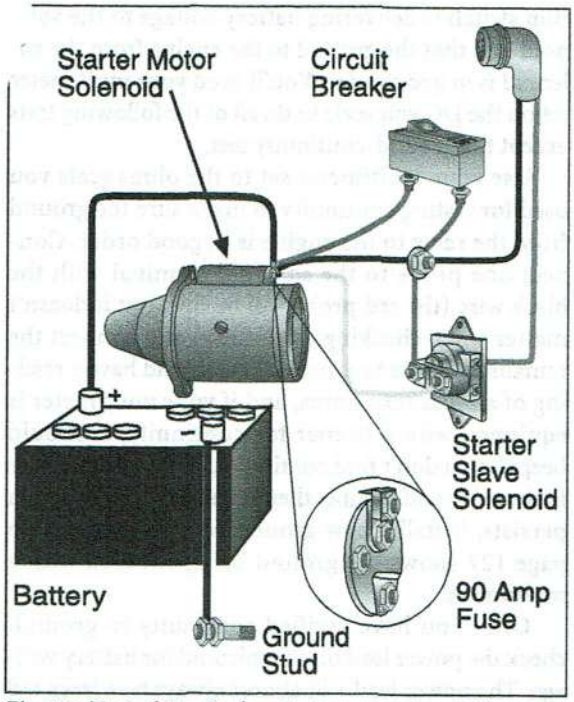


Fig. 8-2. Typical MerCruiser starter-motor circuit diagram.

weakened by a sticking bilge-pump float switch or a light left on while the boat was unattended for an extended period is a common cause of an engine failing to start. Always be certain your battery is charged to at least 70 percent of its capacity before you assume that you have starter-motor problems.

If your boat doesn't have a voltmeter to help you determine battery charge, use your multimeter and take a direct reading of open-circuit voltage at the battery, using the Open-Circut Voltage versus State of Charge table in the Open-Circuit Voltage Test section on page 84. If your battery is low, you will, of course, have to find out why, and the steps outlined in chapter 5 will help you to isolate the cause.

Troubleshooting Starter-Motor Circuits

Once you confirm that the batteries are not the cause of your starter problems, you should begin troubleshooting. Have the workshop manual close at hand so you can identify all the circuit components for your boat.

As a first step, look at all wiring and connections to all starter-circuit parts. Tighten any loose parts and terminals, and clean any corroded terminals. Don't forget to check the fuses and circuit breakers. A blown fuse or a tripped circuit breaker on a starter-motor circuit could be caused by a partially seized starter motor or, in the worst case, a seized engine, both of which situations will be covered in more detail later on in this chapter.

If all connections and fuses or breakers appear to be in good order, a faulty part is certainly the cause for your starter-circuit grief, and a step-by-step approach will be needed to determine which part is at fault.

Starter Solenoid

Most marine inboard engines have a remote *solenoid*, sometimes called a *slave relay*. A solenoid is used as a remote switch to control a circuit, such as your starter circuit, that carries heavy amperage. Acting as a shortcut, the solenoid is connected to the starter switch with a smaller wire to save on the amount of heavy wiring needed to operate the starter circuit. In other words, instead of having a cable as big as your thumb running from the battery to the ignition switch and then from the ignition switch to the starter motor, the heavy cable connects directly to the starter motor through a solenoid. The solenoid is operated by a much smaller (usually 10 or 12 AWG) wire that connects the solenoid to the ignition switch.

Many starter circuits use a solenoid as a remote relay that does not carry full starter-motor current. To determine if your solenoid is intended to carry starter-motor current, first locate the solenoid. It's generally cylindrical and is often found on a bracket at the top forward end of the engine. It will have two large wires and two small wires attached to it. Look at the size of the wire on the solenoid terminals. If the large wires are the same size as your battery cables (typically 4 AWG or larger), the solenoid carries starter-motor current. However, if the large wires are smaller than the battery cables—around the 12, 10, or 8 AWG range—the solenoid does not carry full